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SUBJECT: Prior Art Report for BevoTag: Wi-Fi Fingerprinting, Inertial Fusion, and Item-Tracking Patents That Inform Our Design

INTRODUCTION

This Prior Art Report documents our search of the published research, commercial products, and patents related to BevoTag, and it explains how that search has changed our design. BevoTag is a small tag that locates a laptop, bag, or piece of shared equipment on the University of Texas at Austin campus by matching campus Wi-Fi measurements against a building survey and fusing them with data from a motion sensor. Our Project Definition committed to this approach but left four questions open. We conducted the search to answer them with published evidence before we freeze the system design, and to learn which existing patents limit what we may build.

We reviewed five research publications and five U.S. patents, each selected because it answers at least one research question. The literature shows that channel state information (CSI) reduced error by 30 to 39% relative to signal strength, that our class of microcontroller captures far less CSI when it listens passively, and that fusing Wi-Fi with step counting cut mean error to 1.53 m. The patents show that each of BevoTag's building blocks is already claimed, so our contribution lies in combining them on a self-contained tag. As a result, signal strength becomes our baseline, CSI becomes a measured experiment, and our survey and fusion plans have changed. This report summarizes the design problem, reviews the literature and patents, and explains how the findings answer our research questions.

OVERVIEW OF THE DESIGN PROBLEM

BevoTag must locate a tagged item to the correct room anywhere in surveyed campus buildings and the plazas between them, using only existing campus Wi-Fi and its own motion sensor. The

following subsections summarize the need that motivates the project and the functions and targets that any solution must meet.

Problem Definition and Customer Needs

Students, faculty, and staff need to find misplaced items anywhere on campus, but no existing tracker works reliably indoors without help from other people's phones. Satellite positioning signals do not penetrate buildings well [1, Sec. I]. Consumer tags such as the Apple AirTag and the Samsung Galaxy SmartTag2 update an item's location only when a nearby phone from the same manufacturer detects them [2], [3], so a bag left in an empty lab at night stops reporting. Wi-Fi positioning based on signal strength alone typically errs by 1 to 5 m [4, Table 4], enough to place an item in the wrong room. Our customers therefore need a tag that reports its own location, indoors and on any floor, without new infrastructure and without a passing phone.

Target Functionality and Specifications

Any BevoTag design must measure, transmit, estimate, and display. The tag records the received signal strength indicator (RSSI) and, where the chip supports it, the CSI of nearby access points, and it samples an inertial measurement unit (IMU) that measures acceleration and rotation. A location server compares the Wi-Fi measurements with a survey recorded at known points, a process called fingerprinting, and fuses the result with the IMU data, whose position estimate drifts by more than 150 m within 60 s when it runs alone [5, Sec. 6.3]. A user application then shows the position on a campus map. The governing targets are to place the item in the correct room in more than 90% of trials, keep median error under 5 m, update at least every 10 s while the item moves, and run at least 24 hours on one charge, all at a parts cost of at most $50 per tag. Our prototype uses an ESP32-family microcontroller, which exposes CSI to user firmware [6], and we have built a data logger and a survey application for three buildings, starting with the Engineering Education and Research Center (EER).

REVIEW OF PRIOR ART

We organized our search around four research questions that the Project Definition left open. Each question corresponds to one of our design challenges, and we kept a source only if it gave evidence on at least one question:

1. **Wi-Fi measurement:** Does CSI give enough accuracy beyond RSSI to justify its cost, and can a low-cost chip capture it from campus access points?
2. **Survey effort:** How many survey points, samples, and repeated visits does a building need, and how quickly does a survey go out of date?
3. **Sensor fusion:** How should the tag combine Wi-Fi fixes with IMU data to keep drift bounded, and how much does fusion improve accuracy?
4. **Freedom to operate and novelty:** Which parts of our design do existing products and patents already cover, and where can BevoTag differ?

The first subsection reviews research publications that address questions 1 through 3, and the second reviews patents that address all four. Table 1 on page 6 at the end of the first subsection and Table 2 on page 9 at the end of the second summarize each source against our requirements.

Prior Art Exclusive of Patent Information

Five research publications span the history of Wi-Fi positioning, from the first signal-strength fingerprinting system to CSI capture on the same microcontroller family that our tag uses. Two of them establish RSSI and CSI baselines, one tests our hardware's CSI capture, one measures the benefit of inertial fusion, and one compares many systems in a realistic, changing environment.

Search Procedure

We searched the technical literature with keyword queries that combined "Wi-Fi" or "WLAN" with "fingerprint," "channel state information," "ESP32," "pedestrian dead reckoning," "inertial," and "Kalman" or "particle filter," and we retrieved each candidate's full text from IEEE, ACM, SpringerOpen, or the authors' own pages. We kept papers that report measured accuracy in real buildings, use hardware comparable in cost to ours, or evaluate systems in environments that change, and we confirmed every result we cite against the paper's own tables. We discarded papers that relied only on simulation, that required infrastructure we cannot install, such as ultra-wideband anchors, or that tracked people rather than devices.

RADAR: The First Wi-Fi Fingerprinting System

Bahl and Padmanabhan's RADAR system, presented at IEEE INFOCOM 2000, established the two-phase fingerprinting method that BevoTag adopts [7]. In an off-line phase, the authors recorded signal strength from three base stations at 70 known points on a 43.5 m by 22.5 m office floor with more than 50 rooms; in an on-line phase, they located a laptop by finding the survey points whose signal strengths were nearest in "signal space." This empirical method achieved a median error of 2.94 m, which the authors noted is about the size of an office room, and tracking a walking user raised the median to 3.5 m [7, Secs. 4.1.1, 4.1.7].

RADAR's results guide our survey design more than our accuracy targets. The authors found that about 40 survey points and three real-time samples achieved nearly all of the accuracy of the full data set [7, Secs. 4.1.4, 4.1.5]. When the survey contained only one orientation and the user faced the opposite way, however, the median error rose 67% to 4.90 m, because the user's body changes the signal strength that each base station measures [7, Sec. 4.1.6]. RADAR's median error is close to our 5 m ceiling but not to our room-level target, and its three base stations were dedicated hardware, whereas BevoTag must use whatever campus access points it hears.

DeepFi: CSI Fingerprinting Compared with RSSI

Wang, Gao, Mao, and Pandey's DeepFi system, published in *IEEE Transactions on Vehicular Technology*, provides the clearest comparison of CSI and RSSI fingerprints measured in the same rooms [8]. The authors recorded the amplitudes of 30 subcarriers on each of three antennas of a laptop Wi-Fi card, trained a neural network for each of 50 survey points, and estimated position as a probability-weighted average of the survey points. They compared DeepFi with an earlier CSI method and with Horus, a widely cited probabilistic RSSI system [9], using the same data.

CSI fingerprints were consistently more accurate than RSSI fingerprints. In a mostly empty 4 m by 7 m living room, DeepFi's mean error was 0.94 m and Horus's was 1.54 m; in a cluttered 6 m by 9 m computer laboratory with few line-of-sight paths, the errors were 1.81 m and 2.60 m [8, Table I]. These reductions of 39% and 30% support our plan to evaluate CSI. The study differs from BevoTag in three important ways, however: it used one access point that the authors controlled, a laptop card with three antennas rather than our single-antenna chip, and rooms much

smaller than an EER floor.

ESP32 CSI Toolkit: CSI Capture on a Microcontroller

Hernandez and Bulut showed at IEEE WoWMoM 2020 that the ESP32 microcontroller can capture CSI without the laptops and modified drivers that earlier studies required [10]. Their open-source toolkit runs on a module that costs less than $10, weighs less than 10 g, and reports 64 subcarriers from a single antenna [10, Table I]. The authors used it for activity recognition rather than localization, but they measured the capture rates that matter to us.

The capture rates expose a limit that bears directly on our tag. When the ESP32 was connected to the transmitter, it captured CSI from up to about 600 frames per second; when it only listened, as our tag must to hear many campus access points, it captured just 34.5% as many frames, with greater variation from second to second, because a passive receiver cannot request retransmission [10, Sec. IV-A]. This result confirms that our hardware choice can supply CSI, but it suggests that our tag will collect far fewer CSI samples from campus access points than from a transmitter it controls.

Evennou and Marx: Fusing Wi-Fi with Dead Reckoning

Evennou and Marx of France Telecom fused Wi-Fi fingerprinting with pedestrian dead reckoning in a 2006 *EURASIP Journal on Applied Signal Processing* paper, the closest published precedent for BevoTag's fusion stage [11]. A belt-worn box counted steps with an accelerometer and tracked heading with a gyroscope. The authors compared a Kalman filter with a particle filter that uses the building map to eliminate any position hypothesis that crosses a wall.

Each step of filtering and fusion improved accuracy in their 40 m by 40 m office building with four access points. Along a corridor trajectory, nearest-neighbor fingerprinting gave a mean error of 3.32 m, a statistical fingerprint method 2.88 m, a Kalman filter 2.56 m, the map-aware particle filter 1.86 m, and the particle filter with inertial data 1.53 m [11, Table 1]. The authors also observed that the particle filter's many wall checks make it heavy to compute [11, Sec. 3]. These results support fusion and the use of floor plans, which our survey application already contains, but they were measured with a laptop doing the processing and a survey of only one point per

room.

Microsoft Indoor Localization Competition: Performance Under Change

Lymberopoulos *et al.* reported how 22 indoor positioning systems performed when tested in the same unfamiliar space at the 2014 Microsoft Indoor Localization Competition [12]. Teams were given seven hours to install hardware or survey about 300 m^2 of two rooms and a hallway, and the organizers then rearranged the furniture in one room before measuring each system at 20 test points.

The competition shows that Wi-Fi fingerprinting can rival custom infrastructure but degrades when a room changes. The best system that relied only on existing access points and phone sensors had an average error of 1.6 m, better than most systems that installed their own hardware, and half of all teams achieved less than 3 m [12]. Most teams needed all seven hours to survey or install, and rearranging the furniture raised the error of the leading Wi-Fi systems by up to 0.94 m [12, Table 2]. Because campus rooms change daily as people move chairs and walk through hallways, these findings warn that a one-time survey of EER will not stay accurate.

Table 1. Research Publications Compared with BevoTag Requirements

Source	Question	Key Finding	Difference from BevoTag
RADAR [7]	2	RSSI fingerprinting: 2.94 m median; survey must include orientations	Dedicated base stations; no motion sensing
DeepFi [8]	1	CSI mean error 30–39% below RSSI in the same rooms	One controlled access point; three-antenna laptop card
ESP32 CSI Toolkit [10]	1	$10 chip captures 64-subcarrier CSI; passive rate 34.5% of active	Used for activity sensing, not localization
Evennou and Marx [11]	3	Map-aware particle filter with inertial data: 1.53 m mean error	Laptop processing; one survey point per room
Microsoft competition [12]	2, 3	Best Wi-Fi-only system 1.6 m; furniture changes add up to 0.94 m	Two rooms and a hallway; phones, not tags

Patent Search and Findings

Five U.S. patents cover the survey database, the CSI fingerprint, the fusion step, and the crowdsourced tag model that BevoTag must either build on or avoid. Two have expired, and three remain in force. We describe each patent's claims and compare them with our design; these comparisons are engineering judgments, not legal opinions.

Search Procedure

We searched Google Patents with combinations of "Wi-Fi" or "wireless access point" with "fingerprint," "channel state information," "subcarrier," "dead reckoning," "inertial," "lost and found," and "tag," and we retrieved the full grant documents from the U.S. Patent and Trademark Office. For each candidate, we read the independent claims and recorded the assignee, grant date, and status. We kept the patents whose claims describe a function that BevoTag must perform. Google Patents' status and expiration dates are estimates rather than legal conclusions, so we will confirm the status of any patent that becomes a concern with the Engineering Library staff before we finalize the design.

US 7,414,988: Server for Updating a Location Beacon Database

Skyhook Wireless's patent, granted in 2008 to Jones *et al.*, describes a server that maintains a database of Wi-Fi access point locations across a metropolitan area [13]. Vehicles carrying a GPS receiver and a Wi-Fi radio drive nearly every street in a target area, and the server estimates each access point's position from many readings, discards readings that fall far from a weighted centroid, and updates the database as it discovers new access points. Devices then locate themselves from the positions of the access points they hear. The patent differs from BevoTag in what it stores: it records where each access point is, whereas BevoTag records the signal pattern measured at each survey point inside a building. Google Patents lists the patent's adjusted expiration as September 30, 2026, so its outlier rejection and database updating are now free for us to adopt.

US 9,014,718: Location Sensing Using Channel Fading Fingerprinting

Intel's patent, granted in 2015 to Yang, Yang, and Guo, claims the core of a CSI fingerprinting system [14]. A device scans nearby access points, measures a "channel fading profile" from

selected subcarriers, and finds its location by matching the profile to a database recorded during a survey, using a minimum-distance or maximum-likelihood rule. The dependent claims allow the database to sit on the device or a server, which mirrors our open question of where to compute. This patent is the closest prior art to BevoTag's Wi-Fi stage, and it means that CSI fingerprint matching by itself is not patentable. Google Patents lists it as expired because its maintenance fee was not paid, so we may use the method freely.

US 9,632,169: Using Inertial Navigation to Calibrate an Indoor Positioning System

Ricoh's patent, granted in 2017 to Moraleda and Hull, addresses survey effort [15]. A surveyor walks between known points while inertial navigation tracks the path and Wi-Fi signatures are collected continuously; each signature is then assigned a position along a b-spline fitted to the path, and the surveyor may adjust the spline's control points on a display. This method could survey a floor far faster than stopping at each point, which matters given the competition teams' seven-hour struggle. The patent remains in force until 2035, however, and our survey application currently records scans only at stationary points the surveyor selects, so we will avoid fitting and editing a spline path unless we confirm that our method differs from these claims.

US 9,332,476: Correcting Indoor Positioning Using Wi-Fi Handovers

BlackBerry's patent, granted in 2016 to Shatsky and Kouznetsov, corrects dead-reckoning drift at the moment a device hands over to a new access point [16]. If the dead-reckoning position lies outside a "handover spot" around the new access point, claim 1 resets the position to the nearest point on that spot's border. This patent confirms that drift correction from Wi-Fi is a recognized need, but it corrects only at handovers and only coarsely. BevoTag will instead correct the IMU at every fingerprint fix with a filter, the approach Evennou and Marx published in 2006 [11], and will not reset the estimate to a handover boundary. The patent remains in force until 2034.

US 9,479,920: Power Management in a Crowdsourced Lost-and-Found Service

Apple's patent, granted in 2016 to Mayor, describes the crowdsourced model behind tags like the AirTag [17]. A Bluetooth Low Energy tag broadcasts an identifier; low-power subsystems in a passing phone record that identifier and the Wi-Fi access points the phone hears, each with a time stamp; and the phone's main processor periodically wakes, assigns the tag the stored location of

the access point heard at the same time, and reports it to a server. The patent locates a tag only as precisely as the nearest access point and only when a participating phone is nearby, which are the two weaknesses BevoTag is designed to overcome. Because our tag scans and reports by itself and uses no helper phones, our design does not practice these claims, but the patent's strategy of batching measurements to keep the main processor asleep applies directly to our battery target.

Table 2. Patents Relevant to BevoTag

Patent	Assignee	Relevance to BevoTag	Status (Google Patents)
US 7,414,988 [13]	Skyhook Wireless	Access point database with outlier rejection and updating	Expired Sept. 2026
US 9,014,718 [14]	Intel	CSI fingerprint matching against a survey database	Expired (fee related)
US 9,632,169 [15]	Ricoh	Inertially tracked walking survey fitted with a b-spline	Active to 2035; avoid
US 9,332,476 [16]	BlackBerry	Drift reset to a handover spot border	Active to 2034; avoid
US 9,479,920 [17]	Apple	Crowdsourced Bluetooth tag located by phones	Active to 2035; not practiced

IMPACT OF PRIOR-ART SEARCH ON DESIGN DECISION-MAKING

The search answered each of our four research questions and changed three of our design plans: how we treat CSI, how we survey, and how we fuse. The following subsections answer each question in turn and state the decision that follows.

Wi-Fi Measurement: CSI Is Promising but Unproven on Our Hardware

The literature supports CSI's accuracy but leaves its feasibility on our tag in doubt. DeepFi's 30 to 39% error reduction [8, Table I] is large enough to matter for room identification, but it was measured with three antennas and a cooperating access point. Our single-antenna ESP32, listening passively, should capture only about a third of the CSI frames it could capture from a connected transmitter [10, Sec. IV-A], and campus access points transmit OFDM frames at rates we do not control.

We therefore decided to make RSSI fingerprinting the baseline that every later design must beat

and to evaluate CSI as an addition to it. Our first data logger experiment will record CSI and RSSI together in EER and measure how many usable CSI frames per second the tag captures from campus access points while associated with utexas-iot and while listening passively. If CSI improves room identification on held-out survey points by a margin comparable to DeepFi's, we will keep it; otherwise, we will drop it and save the processing and power. Because Intel's CSI fingerprinting patent has expired [14], either outcome is free of patent risk.

Survey Effort: Orientation and Time Matter More than Density

RADAR and the competition together show that a survey's coverage of conditions matters more than its density. RADAR reached nearly full accuracy with about 40 points, but its median error rose 67% when the survey and the user faced opposite directions [7, Sec. 4.1.6], and the competition's leading Wi-Fi systems lost up to 0.94 m when furniture moved [12, Table 2]. A one-time survey with one heading per point would thus waste much of the effort that the competition showed to be the main cost of fingerprinting.

We have accordingly revised our survey protocol in three ways. At each survey point, we will record scans facing at least four directions, and we will repeat the survey of each benchmark area at three times of day, including class changes, when hallways are crowded. We will also hold out the survey points from one visit for testing, so our accuracy figures reflect a survey that has aged. To keep the database current without full resurveys, we will adapt the outlier rejection and incremental updating of Skyhook's now-expired patent [13]. A walking survey would be faster, but Ricoh's active patent [15] covers the spline-fitted version, so we will continue stationary-point collection for now.

Sensor Fusion: A Map-Aware Filter Bounds Drift

Evennou and Marx's results show that fusion and floor-plan constraints together roughly halve fingerprinting error, from 3.32 m to 1.53 m [11, Table 1], which suggests that our 5 m median target is conservative and that room identification is the harder goal. Their study, the competition's mix of Wi-Fi and inertial entries [12], and BlackBerry's patent [16] all treat Wi-Fi as the source that corrects inertial drift, confirming the architecture in our Project Definition.

Our fusion design now has two candidates rather than an open field. An extended Kalman filter [18] is light enough to run on the tag but cannot easily use walls, whereas a particle filter can discard paths through walls using the floor plans already in our survey application but is heavy to compute [11, Sec. 3]. This trade-off pushes the particle filter toward the server, so the tag will most likely send measurements, not positions. We will compare the two filters on logged EER walks before we decide, and we will correct drift at each fingerprint fix rather than resetting at handovers, avoiding BlackBerry's claims.

Freedom to Operate and Novelty: The Combination Is Our Contribution

The patent search shows that each of BevoTag's building blocks is already public. Fingerprint databases [13], CSI fingerprint matching [14], Wi-Fi drift correction [16], and tag location reporting [17] are all claimed, and the research literature has published Wi-Fi and inertial fusion since at least 2006 [11]. No source we found, however, combines CSI-capable fingerprinting and inertial fusion on a self-contained, battery-powered tag that reports its own position without helper phones or new infrastructure.

We will therefore build on the two expired patents and published methods, avoid the specific claims of the three active patents, and rest any claim of novelty on the integrated tag and its performance on a changing campus rather than on a new positioning algorithm.

CONCLUSION

We searched the prior art to answer four questions about BevoTag's Wi-Fi measurement, survey, fusion, and patent landscape. Five research publications show that CSI fingerprints were 30 to 39% more accurate than RSSI in controlled tests, that our microcontroller captures far fewer CSI frames when it listens passively, that surveys must cover orientations and changing rooms, and that a map-aware fusion filter cut mean error to 1.53 m. Five patents show that every building block of BevoTag is already claimed or published, but two of the most relevant have expired and the other three can be avoided. As a result, RSSI fingerprinting becomes our baseline and CSI an experiment that must earn its place, our survey will record several orientations at several times of day, and our fusion choice narrows to two filters. Our contribution is the integrated, self-reporting tag rather than any single algorithm.

Next, we will use our data logger to measure CSI capture rates and RSSI stability in EER, survey the first benchmark area under the revised protocol, and compare the two filters on logged walks. These measurements will settle where BevoTag computes its position and will feed directly into our system block diagram and component selection.

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